September 7, 2026

# NANCY RUONAN ZHANG


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**The Wharton School, University of Pennsylvania**
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## EMPLOYMENT HISTORY

**Post-doctoral Fellow (Mentored by Terence Speed and Mary Wildermuth)** — **10.2005 - 07.2006**
Departments of Statistics and Plant Biology, University of California Berkeley

**Assistant Professor** — **09.2006 - 06.2011**
Department of Statistics, Stanford University

**Faculty Member** — 09.2006 - 06.2011
Graduate Group in Genomics and Computational Biology
University of Pennsylvania

**Associate Professor** — **07.2011 – 06.2018**
Department of Statistics, The Wharton School, University of Pennsylvania

**Professor** — **07.2018 - Current**
Department of Statistics, The Wharton School, University of Pennsylvania

**Ge Li and Ning Zhao Professor** — **07.2019 - Current**
Department of Statistics, The Wharton School, University of Pennsylvania

**Vice Dean** — **07.2019 - Current**
Wharton Doctoral Programs, The Wharton School, University of Pennsylvania

**Faculty Member** — **07.2024 - Current**
Abramson Cancer Center
Center for Cellular Immunotherapies
Perelman School of Medicine, University of Pennsylvania

## EDUCATION

**Stanford University**

| | |
|---|---|
| **Bachelor's in Mathematics** | **06.2001** |
| **Master's in Computer Science** | **06.2001** |
| **Doctor of Philosophy in Statistics** | **06.2005** |

Dissertation Title: *Change-point models and sequence alignments: Statistical problems of genomics*

## CITIZENSHIP

United States

## HONORS

| | |
|---|---|
| National Defense Science and Engineering Graduate Fellowship | 2002 |
| New World Silver Medal for Best Doctoral Thesis in the Mathematical Sciences | 2007 |
| Stanford University Terman Fellowship | 2006 |
| Sloan Fellowship | 2011 |
| American Statistical Association Medallion Lectureship | 2021 |

| | |
|---|---|
| P.R. Krishnaiah Memorial Lectureship | 2023 |
| Eminent Scholar Award, International Conference on Intelligent Biology and Medicine | 2023 |
| Frontiers of Science Award, International Congress of Basic Science | 2025 |
| David Cox Medal for Statistics | 2025 |

## PUBLICATIONS

### PUBLISHED OR FORTHCOMING IN REFEREED JOURNALS

1. **Zhang NR**, Siegmund DO (2007) A modified Bayes information criterion with applications to the analysis of comparative genomic hybridization data, *Biometrics* 63, 22.
2. Chan HP, **Zhang NR** ‡ (2007) Scan statistics with weighted observations, *Journal of the American Statistical Association*, 102, 595.
3. The ENCODE Project Consortium (2007) Identification and analysis of functional elements in 1% of the human genome by the ENCODE pilot project, *Nature* 447, 799.
4. **Zhang NR**, Wildermuth MC, Speed TP (2008) Transcription factor binding site prediction with multivariate gene expression data, *Annals of Applied Statistics* 2, 332.
5. Lai TL, Xing H, **Zhang NR**‡ (2008) Stochastic segmentation models for array-based comparative genomic hybridization data analysis, *Biostatistics* 9, 290.
6. **Zhang NR**, Senbabaoglu Y, Li J (2010) Joint estimation of DNA copy number from multiple platforms, *Bioinformatics* 26, 153.
7. Siegmund DO, Yakir B, **Zhang NR**‡ (2010) Tail approximations for maxima of random fields by likelihood ratio transformations, *Sequential Analysis* 29, 245.
8. **Zhang NR**, Siegmund DO, Ji H, Li J (2010) Detecting simultaneous changepoints in multiple sequences, *Biometrika* 97, 631.
9. Li F, **Zhang NR**‡ (2010) Bayesian variable selection in structured high-dimensional covariate spaces with applications in genomics, *Journal of the American Statistical Association* 105, 1202.
10. Bickel PJ, Boley N, Brown JB, Huang H, **Zhang NR** ‡ (2010) Subsampling methods for genomic inference, *Annals of Applied Statistics* 4, 1660.
11. Chan HP ¤, **Zhang NR**¤, Chen LHY (2010) Importance sampling of word patterns in DNA and protein sequences, *Journal of Computational Biology* 17, 1697.
12. Chen H, Xing H, **Zhang NR*** (2011) Estimation of parent specific DNA copy number in tumors using high-density genotyping arrays, *PLoS Computational Biology* 7, e1001060.
13. Siegmund DO, Yakir B, **Zhang NR**‡ (2011) Detecting simultaneous variant intervals in aligned sequences, *Annals of Applied Statistics* 5, 645.
14. Efron B and **Zhang NR**‡ (2011) False discovery rates and copy number variation, *Biometrika* 98, 251.
15. Natsoulis G, Bell JM, Xu H, Buenrostro JD, Ordonez H, Grimes S, Newburger D, Jensen M, Zahn JM, **Zhang N**, Ji HP (2011) A flexible approach for highly multiplexed candidate gene targeted resequencing, *PLoS One* 6, e21088.
16. Siegmund DO, **Zhang NR**, Yakir B (2011) False discovery rate for scanning statistics, *Biometrika* 98, 979.
17. Muralidharan O, Natsoulis G, Bell J, Newburger D, Xu H, Keta I, Ji H, **Zhang NR*** (2012) A cross-sample statistical model for SNP detection in short-read sequencing data, *Nucleic Acids Research* 40, e5.

* *corresponding or co-corresponding author*
‡ *alphabetical order*
¤ *co-first authors*

18. Flaherty P, Natsoulis G, Muralidharan O, Winters M, Buenrostro J, Bell J, Brown S, Holodniy M, **Zhang N**, Ji HP (2012) Ultrasensitive detection of rare mutations using next-generation targeted resequencing, *Nucleic Acids Research* 40, e2.
19. Shen J, **Zhang NR**[*] (2012) Change-point model on nonhomogeneous Poisson processes with application in copy number profiling by next-generation DNA sequencing, *Annals of Applied Statistics* 6, 476.
20. Muralidharan O, Natsoulis G, Bell J, Ji H, **Zhang NR*** (2012) Detecting mutations in mixed sample sequencing data using empirical Bayes, *Annals of Applied Statistics* 6, 1047.
21. **Zhang NR**, Siegmund DO (2012) Model selection for high dimensional, multi-sequence change-point problems, *Statistica Sinica* 22, 1507.
22. Sun Y, **Zhang NR** and Owen A (2012) Multiple hypothesis testing, adjusted for latent variables, with an application to the agemap gene expression data, *Annals of Applied Statistics* 6, 1664.
23. Chen H, **Zhang NR**[ǂ] (2013) Graph-based tests for two-sample comparisons of categorical data, *Statistica Sinica* 23, 1479.
24. Natsoulis G, **Zhang NR**, Welch K, Bell J, Ji HP (2013) Identification of insertion deletion mutations from deep targeted resequencing, *Journal of Data Mining in Genomics and Proteomics* 4, 132.
25. Nadauld LD, Garcia S, Natsoulis G, Bell JM, Miotke L, Hopmans ES, Xu H, Pai RK, Palm C, Regan JF, Chen H, Flaherty P, Ootani A, **Zhang NR**, Ford JM, Kuo CJ, Ji HP (2014) Metastatic tumor evolution and organoid modeling implicate TGFBR2 as a cancer driver in diffuse gastric cancer, *Genome Biology* 15,428.
26. Chen H, Bell JM, Zavala NA, Ji HP, **Zhang NR**[*] (2015) Allele-specific copy number profiling by next-generation DNA sequencing, *Nucleic Acids Research* 43, e23.
27. Jiang Y, Oldridge DA, Diskin SJ, **Zhang NR*** (2015) CODEX: a normalization and copy number variation detection method for whole exome sequencing, *Nucleic Acids Research* 43, e39.
28. Chen H, **Zhang NR**[ǂ] (2015) Graph-based change-point detection, *The Annals of Statistics* 43, 139.
29. Cushing A, Kamali A, Winters M, Hopmans ES, Bell JM, Grimes SM, Li CX, **Zhang NR**, Moss RB, Holodniy M, Ji H (2015) Emergence of hemagglutinin mutations during the course of influenza infection, *Scientific Reports* 5, 16178.
30. Peixoto LL, Wimmer ME, Poplawski SG, Tudor JC, Kenworthy CA, Liu S, Mizuno K, Garcia BA, **Zhang NR**, Giese K, Abel T (2015) Memory acquisition and retrieval impact different epigenetic processes that regulate gene expression, *BMC Genomics* 16, S5.
31. Yue M, Han X, De Masi L, Zhu C, Ma X, Zhang J, Wu R, Schmieder R, Kaushik RS, Fraser GP, Zhao S, McDermott PF, Weill FX, Mainil JG, Arze C, Fricke WF, Edwards RA, Brisson D, **Zhang NR**, Rankin SC, Schifferli DM (2015) Allelic variation contributes to bacterial host specificity, *Nature Communications* 6, 8754.
32. Wang X, Chen M, Yu X, Pornputtapong N, Chen H, **Zhang NR**, Powers RS, Krauthammer M (2016) Global copy number profiling of cancer genomes, *Bioinformatics*, 32, 926.
33. **Zhang NR**, Yakir B, Xia LC, Siegmund DO (2016) Scan statistics on Poisson random fields with applications in genomics, *Annals of Applied Statistics* 10, 726.
34. Xia LC, Sakshuwong S, Hopmans ES, Bell JM, Grimes SM, Siegmund DO, Ji HP, **Zhang NR*** (2016) A genome-wide approach for detecting novel insertion-deletion variants of mid-range size, *Nucleic Acids Research* 44, e126.

* *corresponding or co-corresponding author*
ǂ *alphabetical order*
¤ *co-first authors*

35. Jiang Y, Qiu Y, Minn AJ, **Zhang NR*** (2016) Assessing intratumor heterogeneity and tracking longitudinal and spatial clonal evolutionary history by next-generation sequencing, *Proceedings of the National Academy of Sciences* 113, E5528.
36. Wang X, Chen H, **Zhang NR** (2017) DNA copy number profiling using single-cell sequencing, *Briefings in Bioinformatics*, bbx004, https://doi.org/10.1093/bib/bbx004.
37. Jiang Y, **Zhang NR***, Li M* (2017) SCALE: modeling allele-specific gene expression by single-cell RNA-sequencing, *Genome Biology* 18, 74.
38. Chen H, Jiang Y, Maxwell K, Nathanson K, **Zhang NR*** (2017) Allele-specific copy number estimation by whole exome sequencing, *Annals of Applied Statistics* 11, 1169.
39. Jia C, Hu Y, Kelly D, Kim J, Li M*, **Zhang NR*** (2017) Accounting for technical noise in differential expression analysis of single-cell RNA sequencing data, *Nucleic Acids Research*, 45, 10978.
40. Maxwell KN, Wubbenhorst B, Wenz BM, Sloover DD, Pluta J, Emery L, Barrett A, Kraya AA, Anastopoulos IN, Yu S, Jiang Y, Chen H, **Zhang NR**, Hackman N, D'Andrea K, Daber R, Morrissette JJ, Mitra N, Feldman M, Domchek SM, Nathanson KL (2017) BRCA locus-specific loss of heterozygosity in germline BRCA1 and BRCA2 carriers, *Nature Communications* 8, 319.
41. Xia LC, Bell JM, Wood-Bouwens C, Chen JJ, **Zhang NR***, Ji HP* (2017) Single molecule-based discovery of complex genomic rearrangements, *Nucleic Acids Research* 46, e19.
42. Garman B, Anastopoulos IN, Krepler C, Brafford P, Sproesser K, Jiang Y, Wubbenhorst B, Amaravadi R, Bennett J, Beqiri M, Elder D, Flaherty KT, Frederick DT, Gangadhar TC, Guarino M, Hoon D, Karakousis G, Liu Q, Mitra N, Petrelli NJ, Schuchter L, Shannan B, Sheilds CL, Wargo J, Wenz B, Wilson MA, Xiao M, Xu W, Xu X, Yin X, **Zhang NR**, Davies MA, Herlyn M, Nathanson KL (2017) Genetic and genomic characterization of 462 melanoma patient-derived xenografts, tumor biopsies and cell lines, *Cell Reports* 21, 1936.
43. Huang M, Wang J, Torre E, Dueck H, Shaffer S, Bonasio R, Murray J, Raj A, Li M, **Zhang NR*** (2018) SAVER: Gene expression recovery for single cell RNA sequencing, *Nature Methods* 15, 539.
44. Zhou Z, Wang W, Wang L-S, **Zhang NR*** (2018) Integrative DNA copy number detection and genotyping from sequencing and array-based platforms, *Bioinformatics* 34, 2349.
45. Wang X, Jiang Y, **Zhang NR**, Small D (2018) Sensitivity analysis and power for instrumental variable studies, *Biometrics* doi: 10.1111/biom.12873.
46. Urrutia E, Chen H, Zhou Z, **Zhang NR***, Jiang Y* (2018) Integrative pipeline for profiling DNA copy number and inferring tumor phylogeny, *Bioinformatics* 34, 2126.
47. Zhang H, **Zhang NR**, Li M, Reilly MP (2018) First giant steps towards a cell atlas of atherosclerosis, *Circulation Research* 122, 1632.
48. Wang J, Huang M, Torre E, Dueck H, Shaffer S, Murray J, Raj A, Li M, **Zhang NR*** (2018) Gene expression distribution deconvolution in single cell RNA sequencing, *Proceedings of the National Academy of Sciences* 115, E6437.
49. Jiang Y, Nathanson KL, **Zhang NR*** (2018) CODEX2: full-spectrum copy number variation detection by high-throughput DNA sequencing, accepted by *Genome Biology* 19, 202.
50. Wang X, Park J, Susztak K, **Zhang NR***, Li M* (2019) Bulk Tissue Cell Type Deconvolution with Multi-Subject Single-Cell Expression Reference, *Nature Communications* 10, 380.
51. Wang J, Agarwal D, Huang M, Hu G, Zhou Z, Ye C, **Zhang NR*** (2019) Data denoising with transfer learning in single-cell transcriptomics, *Nature Methods* 16, 875.
52. Benci JL et al. (2019) Opposing Functions of Interferon Coordinate Adaptive and Innate Immune Responses to Cancer Immune Checkpoint Blockade, *Cell* 178 (4), 933-948. e14.

* *corresponding or co-corresponding author*
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¤ *co-first authors*

53. Nguyen S et al. (2019) Elite control of HIV is associated with distinct functional and transcriptional signatures in lymphoid tissue CD8+ T cells, *Science Translational Medicine* 11(523).
54. Pauly D, Agarwal D, Dana N, Shafer N, Biber J, Wunderlich KA, Jabri Y, Straub T, **Zhang NR**, Gautam AK, Weber BHF, Hauck SF, Kim M, Curcio CA, Stambolian D, Li M, and Grosche A (2019) Cell-Type-Specific Complement Expression in the Healthy and Diseased Retina. *Cell Reports* 29, 2835-2848. e4
55. Agarwal D, **Zhang NR*** (2019) A rank-based semblance kernel on probability spaces, *Science Advances* 5 (12), eaau9630.
56. Zhou Z, Xu B, Minn A, Zhang NR* (2020) Dendro: genetic heterogeneity profiling by single cell RNA sequencing, Genome Biology 21, 10. (R package: DENDRO)
57. Zhou Z, Ye C, Wang J, **Zhang NR*** (2020) Surface protein imputation from single cell transcriptomes by deep neural networks, *Nature Communications* 11, Article number: 651
58. Agarwal D, Wang J, **Zhang NR*** (2020) Data denoising and post-denoising corrections in single cell RNA sequencing, Statistical Science 35 (1), 112-128.
59. Rozenblatt-Rosen et al. (2020) The Human Tumor Atlas Network (HTAN): charting tumor transitions across space and time at single-cell resolution. Cell 181, 236.
60. Mukherjee S, Agarwal D, **Zhang NR** & Bhattacharya BB (2020) Distribution-Free Multisample Tests Based on Optimal Matchings with Applications to Single Cell Genomics, Journal of the American Statistical Association, DOI: 10.1080/01621459.2020.1791131.
61. Wu C-Y, Lau BT, Kim H, Sathe A, Grimes SM, Ji HP, **Zhang NR*** (2021) Integrative single-cell analysis of allele-specific copy number alterations and chromatin accessibility in cancer. *Nature Biotechnology,* 39, 1259
62. Navin NE, Rozenblatt-Rosen O, and **Zhang NR** (2021) New frontiers in single-cell genomics. *Genome Research*, 31, ix-x.
63. Zhao Q*, Wang J, Miao Z, **Zhang NR**, Hennessy S, Small DS, Rader DJ (2021) A Mendelian randomization study of the role of lipoprotein subfractions in coronary artery disease. *eLife 10, e58361*.
64. Wang J, Zhao Q, Bowden J, Hemani G, Smith GD, Small DS, **Zhang NR** (2021) Causal inference for heritable phenotypic risk factors using heterogeneous genetic instruments. *PLoS Genetics* 17(6), e1009575.
65. Cucolo L, …, **Zhang NR**, Shi J and Minn AJ (2022). The interferon-stimulated gene RIPK1 regulates cancer cell intrinsic and extrinsic resistance to immune checkpoint blockade, *Immunity* 55, 671.
66. Bibby J, Agarwal D, Freiwald T, Kunz N, Merle NS, West EE, Larochelle A, Chinian F, Mukherjee S, Afzali B, Claudia K*, and **Zhang NR*** (2022). Systematic Single Cell Pathway Analysis (SCPA) reveals novel pathways engaged during early T cell activation. *Cell Reports* 41, 111697*.*
67. Jiang Y, Harigaya Y, Zhang Z, Zhang H, Zang C, and **Zhang NR*** (2022) Nonparametric single-cell multiomic characterization of trio relationships between transcription factors, target genes, and cis-regulatory regions. *Cell Systems* 13, 737*.*
68. Sathe A, …, **Zhang NR**, Ji HP (2022) Colorectal cancer metastases in the liver establish immunosuppressive spatial networking between tumor associated SPP1+ macrophages and fibroblasts. Clinical Cancer Research OF1–OF1.
69. Hickey JW, …, **Zhang NR**, … (2023) Organization of the human intestine at single-cell resolution. Nature 619 (7970), 572-584.
70. Lin K and **Zhang NR*** (2023) Quantifying common and distinct information in single-cell multimodal data with Tilted Canonical Correlation Analysis. *Proceedings of the National Academy of Sciences, 120 (32) e2303647120.*


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ǂ *alphabetical order*
¤ *co-first authors*

71. Chen S, Zhu B, Huang S, Hickey JW, Lin KZ, Snyder M, Greenleaf WJ, Nolan GP, **Zhang NR***, Ma Z* (2024) Integration of spatial and single-cell data across modalities with weak linkage. *Nature Biotechnology* 42, 1096–1106
72. Mason K, Sathe A, Hess P, Rong J, Wu C-Y, Furth E, Susztak K, Levinsohn J, Ji HP, **Zhang NR*** (2024) Niche-DE: niche differential gene expression analysis in spatial transcriptomics data identifies context-dependent cell-cell interactions. Genome Biol 25, 14.
73. Mathew D. et al. (2024) Clinical and Immunological Responses to Combined JAK Inhibition and PD1 Blockade for Lung Cancer. *Science, 384,* 1303-1304.
74. Zhang Z, Mathew D, Lim TL, Mason K, Martinez CM, Huang S, Wherry EJ, Susztak K, Minn AJ, Ma Z*, **Zhang NR*** (2024) Recovery of biological signals lost in single-cell batch integration with CellANOVA. *Nature Biotechnology 43, 1861-1877*.
75. Xu J¤, Chen C¤, Sussman JH, Yoshimura S, Vincent T, Pölönen P, …, **Zhang NR**, …, Mullighan CG, Tan K*, Teachey DT* (2024) A multiomic atlas identifies a treatment-resistant, bone marrow progenitor-like cell population in T cell acute lymphoblastic leukemia. *Nature Cancer 6, 102-122*.
76. Malamon JS¤, Farrell JJ¤, Xia LC, Dombroski BA, Das RG, Way J, …, **Zhang NR**, Wang L-S, Farrer LA, Schellenberg GD, Lee WP¤*, Vardarajan BN¤* (2024) A comparative study of structural variant calling in WGS from Alzheimer's disease families. *Life Science Alliance 7, e202302181*.
77. Wu C-Y, Rong J, Sathe A, Hess PR, Lau BT, Grimes SM, Huang S, Ji HP*, **Zhang NR*** (2025) Cancer subclone detection based on DNA copy number in single-cell and spatial omic sequencing data. *Nature Methods 22, 1846-1856*.
78. Yu W, Biyik-Sit R, Uzun Y, Chen C-H, Thadi A, Sussman JH, Pang M, Wu C-Y, Grossmann LD, Gao P, Wu DW, Yousey A, Zhang M, Turn CS, Zhang Z, Bandyopadhyay S, Huang J, Patel T, Chen C, Martinez D, Surrey LF, Hogarty MD, Bernt K, **Zhang NR**, Maris JM, Tan K (2025) Longitudinal single-cell multiomic atlas of high-risk neuroblastoma reveals chemotherapy-induced tumor microenvironment rewiring. *Nature Genetics 57, 1142-1154*.
79. Klötzer KA, Abedini A, Li S, Balzer MS, Liang X, Levinsohn J, Ha E, Hogan JJ, Quinn G, Bloom RD, Schuller M, Eller K, Halmos B, **Zhang NR***, Susztak K* (2025) Analysis of individual patient pathway coordination in a cross-species single-cell kidney atlas. *Nature Genetics 57, 1922-1934*.
80. Li H, Wang D, Gao Q, Tan P, Wang Y, Cai X, Li A, Zhao Y, Thurman AL, Malekpour SA, Zhang Y, Sala R, Cipriano A, Wei C-L, Sebastiano V, Song C, **Zhang NR**, Au KF* (2025) Improving gene isoform quantification with miniQuant. *Nature Biotechnology 44, 477-489*.
81. Fu Y, Kim H, Roy S, Huang S, Adams JI, Grimes SM, Lau BT, Sathe A, Ji HP, **Zhang NR*** (2025) Single cell and spatial alternative splicing analysis with Nanopore long read sequencing. *Nature Communications 16, 6654*.
82. Gier RA, Bracht SA, Rong J, Reyes Hueros RLA, Wahlsten ML, Cote C, DeMarshall M, Karakasheva TA, Strauss Starling A, Muir AB, Falk GW, **Zhang NR**, Shaffer SM* (2025) Clonal cell states link gastroesophageal junction tissues with metaplasia and cancer. *Nature Communications 16, 10952*.
83. Lin Y, Wang H, Wilson PC, **Zhang NR*** (2026) Robust footprinting with sample-specific Tn5 bias correction for bulk and single cell ATAC-seq. *Nature Communications 17*.
84. Enninful A¤, Zhang Z¤, Klymyshyn D, Ingalls M, Yang M, Zong H, Bai Z, Farzad N, Su G, Baysoy A, Nam J, Lu Y, Bao S, Deng S, **Zhang NR**, Braubach O, Xu ML*, Ma Z*, Fan R* (2026) Integration of imaging-based and sequencing-based spatial omics mapping on the same tissue section via DBiTplus. *Nature Methods 23, 1827-1839*.
85. Schaff DL, White PE, Cote CJ, Watterson GE, Lin KZ, Fasse AJ, **Zhang NR**, Shaffer SM* (2026) Pre-existing cell states predict resistance to multiple treatments. *Cell Genomics 6, 101191*.

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86. Sussman JH¤, Oldridge DA¤, Yu W¤, Chen C-H, Zellmer AM, Rong J, …, Viaene AN, **Zhang NR**, De Raedt T, Cole K, Tan K* (2026) A longitudinal single-cell and spatial multiomic atlas of pediatric high-grade glioma. *Cell Reports Medicine 7, 102766*.
87. Mason K, Jiang Y, Klotzer KA, **Zhang NR*** (2025) Ultra-scalable differential testing for spatial omics. *Submitted.*
88. Chen XE, Lin KZ, Schaff D, Vander Velde R, Cote C, Huang S, Minn AJ, Shaffer SM, **Zhang NR*** (2025) Temporal and Clonal Resolution of Cellular Evolution Under Stress, *Submitted*.
89. Yang Y, Hess PR, Huang S, Teneche MG, Wang H, Miller KN, Davis AE, Miciano C, Li KY, Mamde S, Yip K, Ren B, Yang Q, Smoot E, Wang A, Johnson B, Wilson P, Adams PD, **Zhang NR*** (2026) A measure of transcriptional dyscoordination for quantifying aging in single cells. *bioRxiv, submitted*.
90. Fu Y, Mathew D, Wang M, Huang S, Chen XE, Lin KZ, Schaff, Shaffer SM, Pardoll DM, Jackson C, Zhang NR (2026) Deciphering cell fate and clonal dynamics via integrative single-cell lineage modeling. *Submitted.*
91. Bartolo L, Afroze S, Lin Y, Ansari A, Pan Y-G, Liu C, **Zhang NR**, Naji A, Su LF* (2025) Intestinal Immune Dysregulation Fuels Islet Autoimmunity in Type 1 Diabetes, *Submitted*.
92. Qiu J, Ye D, Chen XE, Dangle N, Yoshor B, Zhang T, Shao Y, **Zhang NR**, Minn AJ* (2024) Targeting interferon-driven inflammatory memory prevents epigenetic evolution of cancer immunotherapy resistance. *bioRxiv, submitted*.
93. Bracht SA, Rong J, Gier RA, DeMarshall M, Golden H, Dhakal D, **Zhang NR**, Shaffer SM* (2025) Mitochondrial clone tracing within spatially intact human tissues. *bioRxiv, submitted*.
94. Wang M, Fu Y, Bom S, Ning Y, Matthews D, Zhang M, Lucas CH, Choi J, **Zhang NR**, Jackson CM* (2025) Dual checkpoint blockade of glioblastoma with anti-PD-1 and anti-LAG-3 promotes expansion of tumor-reactive T cell clones along a unique pathway of differentiation. *bioRxiv, submitted*.
95. Miao Z, Qu Y, Huang S, Laux L, Peters S, Aristel A, Zhang Z, **Zhang NR*** (2026) Dissecting the coordinated progression of cell states in spatial transcriptomics with CoPro. *bioRxiv, submitted*.


PUBLISHED BOOK CHAPTERS


96. Chan HP, Tu I-P, **Zhang NR** (2009) Boundary crossing probability computations in the analysis of scan statistics, in *Scan Statistics - Methods and Applications.* Birkhauser, Boston.
97. **Zhang NR** (2010) DNA copy number profiling in normal and tumor genomes, In *Frontiers in Computational and Systems Biology*, ed. Jianfeng Feng, Wenjiang Fu and Fengzhu Sun.


## GRANT AWARDS

*(**NSF** = National science foundation, **DMS** = Division of Mathematical Sciences, **NIH** = National Institutes of Health, **NHGRI** = National Human Genome Research Institute, **DoJ**=Department of Justice, **NIA** = National Institutes of Aging, **NCI** = National Cancer Institute, **NHLBI** = National Heart, Lung and Blood Institute)*

| Period | Agency, *Mechanism* | Role | Title | Direct Cost ($) |
|---|---|---|---|---|
| 2009 - 2012 | NSF (DMS) | PI | Change-point Problems in Genomic Profiling | 100,000 |
| 2010-2013 | NSF (DMS) | co-PI | Statistical Methods for Threat Detection | 711 |

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ǂ *alphabetical order*
¤ *co-first authors*

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|---|---|---|---|---|
| 7/6/11 - 6/30/17 | NIH (NHGRI) *R01* | PI | Statistical Models and Analysis of Complex Variation in Clonal Mixtures | 577,971 |
| 9/15/11 - 9/15/13 | Alfred P. Sloan Foundation | PI | Statistical Methods for Genome Profiling | 50,000 |
| 5/1/12 - 4/30/14 | NIH (NHGRI) *R01* | PI (Subcontract) | Statistical Models for Genome Sequencing and Association | 81,110 |
| 1/1/14 - 12/31/16 | DoJ | PI (Subcontract) | Highly Parallel Analysis of Complex Genetic Mixtures | 72,731 |
| 6/15/14 – 5/31/18 | NIH (NIA) *UF1* | Co-Investigator | Consortium for Alzheimers Sequence Analysis (CASA) | 11,063,917 |
| 4/15/16 – 2/28/21 | NIH (NIA) *U54* | Co-Investigator | Coordinating Center for Genetics and Genomics of Alzheimers Disease (CGAD) | 10,801,796 |
| 5/1/16 – 4/30/20 | NSF (DMS) | Co-PI | Statistical Methods for High- Resolution Multiscale Analysis 3D DNA | 948,742 |
| 9/30/16 – 6/30/21 | NIH *U54* | Co-Investigator | Identifying Genes and Pathways that Impact Tau Toxicity in FTD | 1,250,000 |
| 4/1/17 – 3/31/22 | NIH (NIA) *U24* | Co-Investigator | The NIA Genetics of Alzheimer's Disease Data Storage Site (NIAGADS) | 4,802,337 |
| 8/1/17- 7/31/22 | NIH (NCI) *P01* | Co-Investigator | Radiation and Checkpoint Blockade for Cancer Immune Therapy | 8,795,373 |
| 9/1/17 – 8/31/21 | NIH (NIGMS) *R01* | PI (Multiple PI Grant) | Statistical Methods for Single- Cell Transcriptomics | 948,000 |
| 9/14/17 – 6/30/20 | NIH (NHGRI) *R01* | PI (Multiple PI Grant) | Genomic and Cellular Variation from Single Molecules to Single Cells | 917,539 |
| 7/1/18- 6/30/23 | NIH (NHLBI) *R01* | Co-Investigator | Elucidation of Tissue-Specific Transcriptomic Profiles in Cardiometabolic Disease | 2,222,228 |
| 9/01/18- 8/31/23 | NIH (NCI) *U2C* | Data Analysis Unit Co-lead | Center for Pediatric Tumor Cell Atlas | 13,553,635 |
| 7/1/23- 6/30/26 | NSF (DMS) | PI | Statistical Methods for Design and Analysis of Clinical-scale Single Cell Studies | 600,000 |
| 9/01/18- 8/31/23 | NIH (NIGMS) *R01* | PI (Multiple PI Grant) | Multiomic single cell and spatial interrogation of mechanisms in cellular adaptation to stress | 1,772,212 |
| 9/2024- 2025 | NIH (NIA) R56 | PI (Multiple PI Grant) | Multiomic methods for the characterization of cellular aging | 250,000 |

## COURSES TAUGHT

| | |
|---|---|
| Stanford University STATISTICS 191 – Applied Statistics | 2007, 2008 |
| Stanford University STATISTICS 203 – Introduction to ANOVA | 2009, 2010 |
| Stanford University STATISTICS 205 – Nonparametric Statistics | 2007, 2008 |
| Stanford University STATISTICS 215 – Stochastics Processes with Applications in Biology | 2008, 2009, 2010 |
| Stanford University STATISTICS 345/GEN245 – Computational Algorithms in Statistical Genetics | 2009 |
| Stanford University STATISTICS 366 – Statistical Methods in Genetics | 2010 |
| Wharton School STAT 102 – Introductory to Business Statistics | 2012, 2015 |
| Wharton School STAT 431 – Introductory Statistics | 2012 |
| Wharton School STAT 471/701 – Intermediate Statistics | 2013 |
| Wharton School STAT 405/705 – Statistical Computing with R | 2016, 2017, 2019 |
| Wharton School STAT 9915 – Advanced Topics in Biomedical Data Science | 2025 |

* *corresponding or co-corresponding author*
ǂ *alphabetical order*
¤ *co-first authors*

## MENTORING

*I served (or am serving) as doctoral dissertation advisor for:*

| | |
|---|---|
| **Yunting Sun** (Joint with Art Owen), Department of Statistics, Stanford University (Joined Google Inc.) | 2012 |
| **Jeremy Shen**, Department of Statistics, Stanford University (Joined Two Sigma Investments.) | 2012 |
| **Hao Chen** (Joint with David Siegmund), Department of Statistics, Stanford University (Joined Department of Statistics as Assistant Professor, University of California Davis.) | 2014 |
| **Yuchao Jiang**, Graduate Program in Genomics and Computational Biology, University of Pennsylvania (Joined Departments of Biostatistics and Genetics as Assistant Professor, University of North Carolina.) | 2017 |
| **Yang Jiang** (Joint with Dylan Small) Department of Statistics, University of Pennsylvania | 2017 |
| **Xuran Wang**, Graduate Program in Applied Mathematics and Computational Sciences, University of Pennsylvania (Joined Carnegie Mellon University as Postdoctoral Researcher) | 2019 |
| **Mo Huang**, Department of Statistics, The Wharton School, University of Pennsylvania (Joined Merck Pharmaceuticals) | 2020 |
| **Zilu Zhou**, Graduate Program in Genomics and Computational Biology, University of Pennsylvania (Joined Google Research) | 2020 |
| **Divyansh Agarwal**, Graduate Program in Genomics and Computational Biology, University of Pennsylvania (Joined Massachusetts General Hospital, Harvard Medical School as Resident.) | 2020 |
| **Chi-Yun Wu**, Graduate Program in Genomics and Computational Biology, University of Pennsylvania (Joined Gladstone Institute as Postdoctoral Researcher.) | 2022 |
| **Kaishu Mason,** Department of Data Science, University of Pennsylvania (Joined the New York Mets as Senior Data Scientist.) | 2025 |
| **Yuntian Fu,** Graduate Program in Genomics and Computational Biology, University of Pennsylvania (Joined Dana-Farber Cancer Institute as Postdoctoral Researcher.) | 2026 |
| **Zhaojun Zhang**, Department of Statistics and Data Science, University of Pennsylvania (Joined Citadel as Quantitative Researcher.) | 2026 |
| **Jiazhen Rong**, Graduate Program in Genomics and Computational Biology, University of Pennsylvania | Current |
| **Yuxuan Lin**, Department of Statistics and Data Science, University of Pennsylvania | Current |
| **Emilia Chen**, Graduate Program in Genomics and Computational Biology, University of Pennsylvania | Current |
| **Yilin Yang**, Department of Statistics and Data Science, University of Pennsylvania | Current |
| **Yijia Jiang,** Graduate Program in Genomics and Computational Biology, University of Pennsylvania | Current |
| | Current |

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ǂ *alphabetical order*
¤ *co-first authors*

| | |
|---|---|
| **Tracy Wang,** Graduate Program in Genomics and Computational Biology, University of Pennsylvania | Current |
| **Nakial Cross**, Graduate Program in Genomics and Computational Biology, University of Pennsylvania | |

*I served (or am serving) as postdoc mentor for:*

| | |
|---|---|
| **Charlie Xia** (Joint with Hanlee Ji, joined Albert Einstein College of Medicine as Assistant Professor.) | 2017 |
| **Jingshu Wang** (Joined University of Chicago Department of Statistics as Assistant Professor.) | 2019 |
| **Kevin Lin** (Joined University of Washington Department of Biostatistics as Assistant Professor) | 2023 |
| **Paul Hess** | 2024 |
| **Divij Mathews** (co-mentor with Dr. John Wherry, joined Department of Immunology and Microbiology, University of Colorado) | 2025 |

*I served (or am serving) as Master Thesis mentor for:*

| | |
|---|---|
| | 2024 |
| **Jenea Adams** | 2025 |
| **Zhen Miao** (Joined Fred Hutchinson Cancer Center as Postdoctoral Researcher.) | 2026 |
| **Alisha Aristel** (Joined the National Institutes of Health as Research Fellow.) | 2026 |

*Since starting at Penn in 2011, I've served on the Thesis Advising Committees of:*

| | |
|---|---|
| **Jun Chen**, Graduate Group in Genomics and Computational Biology | 2012 |
| **Jonathan Toung**, Graduate Group in Genomics and Computational Biology | 2013 |
| **Joseph Glassner**, Graduate Group in Genomics and Computational Biology | 2014 |
| **Vicky Wu**, Department of Biostatistics, Epidemiology and Informatics | 2014 |
| **Scott Sherrill-Mix**, Graduate Group in Genomics and Computational Biology | 2015 |
| **Hannah Dueck**, Graduate Group in Genomics and Computational Biology | 2015 |
| **Yih-Chii Hwang**, Graduate Group in Genomics and Computational Biology | 2015 |
| **Hyunseung Kang**, Department of Statistics | 2017 |
| **Ying Chen**, Graduate Group in Genomics and Computational Biology | 2017 |
| **Xiao Ji**, Graduate Group in Genomics and Computational Biology | 2017 |
| **Xinyao Ji**, Department of Statistics | 2017 |
| **Cheng Jia,** Department of Biostatistics, Epidemiology and Informatics | 2017 |
| **Yu Hu,** Department of Biostatistics, Epidemiology and Informatics | 2018 |
| **Gemma Moran,** Department of Statistics | 2019 |
| **Benjamin Emert**, Graduate Group in Genomics and Computational Biology | 2021 |
| **Katerina Gawronski**, Graduate Group in Genomics and Computational Biology | 2021 |
| **Gregory Way**, Graduate Group in Genomics and Computational Biology | 2019 |
| **Eric Sanford**, Graduate Group in Genomics and Computational Biology | 2021 |
| **Sammy Klasfeld**, Graduate Group in Genomics and Computational Biology | 2021 |
| **Jingya Qiu,** Graduate Group in Genomics and Computational Biology | 2022 |
| **Yang Xu,** Graduate Group in Genomics and Computational Biology | 2023 |

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| | |
|---|---|
| **Jason Xu,** Graduate Group in Genomics and Computational Biology | 2023 |
| **Sagnik Nandy,** Department of Statistics | 2023 |
| **Kathy Huang**, Graduate Group in Genomics and Computational Biology | 2024 |
| **Michelle Lee**, Graduate Group in Genomics and Computational Biology | 2024 |
| **Zhen Miao**, Graduate Group in Genomics and Computational Biology | 2024 |
| **Matthew Lee**, Graduate Group in Genomics and Computational Biology | Current |
| **Alice Wang,** Graduate Group in Genomics and Computational Biology | Current |
| **Robert Hu**, Graduate Group in Genomics and Computational Biology | Current |
| **Zirui Fan**, Department of Statistics and Data Science | Current |
| **Sydney Bracht**, Department of Bioengineering | Current |
| **Zhuoran Xu,** Graduate Group in Genomics and Computational Biology | Current |
| **Barbara Xiong,** Graduate Group in Genomics and Computational Biology | Current |
| **Ziang Niu**, Department of Statistics and Data Science | Current |
| **Gary Wang**, Department of Bioengineering | Current |
| **Seung Yub Han**, Department of Biology | Current |
| **David Wu,** Graduate Group in Genomics and Computational Biology | Current |
| **Joseph Louis Deutsch**, Department of Statistics | Current |
| **Stephen Phan**, Department of Bioengineering | Current |
| **Minh Nguyen,** Graduate Group in Genetics and Epigenetics | Current |
| **Ijeoma Meremikwu,** Graduate Group in Genomics and Computational Biology | Current |

## *SELECT* NOTABLE SERVICE ACTIVITIES

### *NOTABLE EDITORIAL SERVICE*

| | |
|---|---|
| Associate Editor, *Annals of Applied Statistics* | 2015-2018 |
| Editorial Board, *Briefings in Bioinformatics* | 2017-2021 |
| Guest Editor, *Genome Research Special Issue on Single Cell Biology* | 2020-2021 |
| Editor, *Foundations and Trends in Statistics* | 2024-Current |

### *GRANT REVIEW PANEL AND STUDY SECTIONS*

| | |
|---|---|
| National Science Foundation – National Institute of General Medical Sciences Joint Study Section | 2011 |
| National Institutes of Health – Genomics, Computational Biology and Technology (GCAT) | 2012, 2015, 2017 |
| National Institutes of Health – Advanced Genomic Technology Development Panel | 2017 |
| National Institutes of Health – Advanced Genomic Technology Development Panel | 2021 |
| National Institutes of Health – ASPA | 2022 |
| National Institutes of Health – BICAN | 2022 |
| National Institutes of Health – ZRG1 BST-J | 2022 |
| National Institutes of Health – HTAN, Pre-Cancer Atlas Research Centers | 2024 |

### *Notable ACADEMIC SERVICE*

| | |
|---|---|
| Chair, American Statistical Association Section on Statistical Genetics and Genomics | 2023-2024 |
| Chair, Founder’s Award Committee, ASA SSGG | 2023-2024 |

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ǂ *alphabetical order*
¤ *co-first authors*

*(At Stanford)*

| | |
|---|---|
| Masters student advisor, Department of Statistics | 2010 |
| Undergraduate advisor, Computational Mathematics Major | 2007-2011 |
| VPUE Undergraduate Summer Research Program Coordinator | 2007 |

*(At Penn)*

| | |
|---|---|
| Doctoral Program Co-Director, Department of Statistics, The Wharton School | 2012-2017 |
| Doctoral Program Advisory Committee, Graduate Group in Genomics and Computational Biology | 2014-2019 |
| Center for Neurodegeneration Faculty Search Committee | 2016-2017 |
| Director of Admissions, Department of Statistics, The Wharton School | 2017-2019 |
| Founder, Wharton Directed Reading Program | 2021 |
| Chair, Doctoral Coordinating Committee, Department of Statistics and Data Science | 2023-current |
| Co-Director, Bridge to Doctorate Program, Department of Statistics and Data Science | 2023-current |

## *SELECT* INVITED TALKS (SINCE 2013)

| | |
|---|---|
| Department of Statistics, Stanford University | 2013 |
| Department of Statistics, Harvard University | 2013 |
| Invited Lecture, IMS-China Meeting, Chengdu, China | 2013 |
| Department of Biostatistics, Johns Hopkins University | 2014 |
| Invited Lecture, ENAR Spring Meeting, Baltimore, MD | 2014 |
| Invited Lecture, iBright Conference, Houston, TX | 2015 |
| Department of Statistics, Georgia Institute of Technology | 2016 |
| Invited Lecture, Center for Statistics and Machine Learning, Princeton University | 2016 |
| Invited Lecture, Cornell Day of Statistics, Ithaca, NY | 2016 |
| Invited Lecture, ICSA Applied Statistics Symposium, Atlanta, GA | 2016 |
| Department of Biostatistics, Brown University | 2016 |
| Department of Statistics, Stanford University | 2016 |
| Invited Lecture, Graybill Conference, Fort Collins, CO | 2017 |
| Department of Biostatistics, University of Michigan | 2017 |
| Invited Lecture, ENAR Spring Meeting, Washington, DC | 2017 |
| Invited Lecture, Joint Statistical Meetings, Baltimore, MD | 2017 |
| ICSA Applied Statistics Symposium, Chicago, IL | 2017 |
| Department of Statistics, Pennsylvania State University | 2017 |
| Machine Learning Seminar Series, Duke University | 2017 |
| Invited Lecture, DahShu Virtual Journal Club | 2017 |
| Invited Lecture, Biostatistics Branch, National Cancer Institute | 2018 |
| Invited Lecture, ENAR Spring Meeting, Atlanta, GA | 2018 |
| Invited Lecture, Joint Statistical Meetings, Vancouver, Canada | 2018 |
| Department of Statistics, University of Chicago | 2018 |
| Department of Biostatistics, University of Washington | 2018 |
| Department of Statistics and Data Science, Carnegie Mellon University | 2018 |
| Keynote Lecture, The Australian Bioinformatics and Computational Biology Society Annual Conference | 2018 |
| Department of Biostatistics, University of North Carolina | 2019 |

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¤ *co-first authors*

| | |
|---|---|
| Frontiers in Single-cell Technology, Applications and Data Analysis (Banff Workshop) | 2019 |
| Invited Lecture, New York University Genomics Symposium | 2019 |
| Mathematical Biosciences Institute, Ohio State University | 2019 |
| Invited Lecture, Institute for Advanced Studies (Missing Data Challenges in Computation, Statistics, and Applications), Princeton University | 2020 |
| Department of Statistics, Yale University | 2020 |
| Department of Statistics, Stanford University | 2020 |
| Department of Statistics, UC Berkeley | 2020 |
| Invited Lecture, Joint Statistical Meetings | 2020 |
| Invited Lecture, Gordon Conference on Single Cell Cancer Biology | 2020 |
| Invited Lecture, Keystone Symposia on Single Cell Biology | 2021 |
| Invited Lecture, American Society of Nephrology Kidney Week | 2021 |
| Medallion Lecture, Joint Statistical Meetings | 2021 |
| Invited Lecture, Bioinformatics Symposium (University of Pittsburgh) | 2021 |
| Invited Lecture, Moffitt Cancer Center Quantitative Science Grand Rounds | 2021 |
| Invited Lecture, NeurIPS Learning Meaningful Representations of Life | 2021 |
| Invited Lecture, Princeton Day of Statistics | 2021 |
| Department of Statistics and Biostatistics, University of Calgary | 2021 |
| Keynote Lecture, Conference on Statistics and Data Science | 2021 |
| Invited Lecture, Gordon Research Conference in Single-cell Cancer Biology | 2022 |
| Invited Lecture, Simons Institute Conference - Statistics in the Big Data Era | 2022 |
| Invited Lecture, Penn Biomedical Data Science Seminar | 2022 |
| Invited Lecture, Joint Statistical Meetings, Washington D.C. | 2022 |
| Invited Lecture, BIRS Workshop in Deep Learning in Genomics | 2022 |
| Invited Lecture, NCI Joint Workshop on Computational Approaches to Immuno-Oncology | 2022<br>2023 |
| Stanford University Department of Biomedical Data Science Seminar | 2023 |
| Johns Hopkins Department of Biostatistics Seminar | 2023 |
| University of Michigan Statistics Seminar | 2023 |
| Krishnaiah Lecture, Penn State Department of Statistics | 2023 |
| University of Michigan Biostatistics Seminar | 2023 |
| Keynote Lecture, 11th International Conference on Intelligenet Biology and Medicine | 2023 |
| Keynote Lecture, Bioconductor Conference | 2023 |
| Invited Lecture, Stanford Conference on Genetics, Structural Variants and DNA Repeats | 2023 |
| Mount Sinai Department of Genetics and Genomic Sciences | 2023 |
| Columbia University Department of Statistics | 2024 |
| Emory University Rollins School of Public Health Department of Biostatistics | 2024 |
| University of California, Los Angeles, Frontiers in Bioinformatics Series | 2024 |
| University of Southern California, Department of Statistics | 2024 |
| Invited Lecture, ENAR Spring Meeting, Baltimore, MD | 2024 |
| Invited Lecture, STATGEN 2024, Pittsburgh, PA | 2024 |
| Invited Lecture, National Institute on Aging Workshop on Heterogeneity of Aging: New Perspectives | 2024<br>2024 |
| Invited Lecture, Joint Statistical Meetings, Portland OR | 2024 |
| Invited Lecture, Penn Big Data Conference | 2024 |
| University of Pittsburgh, Department of Statistics | 2024 |

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¤ *co-first authors*

| | |
|---|---|
| University of Illinois at Urbana Champagne, Department of Statistics | 2024 |
| Invited Lecture, 2024 Conference on Molecular and Cellular Aging | 2024 |
| Keynote Lecture, Lange Symposium, University of California at Los Angeles | 2025 |
| Harvard University, Department of Statistics | 2025 |
| Harvard University, Department of Biostatistics | 2025 |
| ENAR 2025 Spring Meeting New Orleans | 2025 |

## *SELECT* SOFTWARE PACKAGES DEVELOPED BY MY GROUP

*For single cell data analysis:*

TASC (Toolkit for noise modeling in single cell RNA-seq with spike-ins)
https://github.com/scrna-seq/TASC
SCALE (Single cell allele-specific expression analysis)
https://github.com/yuchaojiang/SCALE
DESCEND (Expression distribution deconvolution for single cell RNA-seq)
https://github.com/jingshuw/descend
MUSIC (Bulk expression deconvolution with scRNA-seq reference)
https://github.com/xuranw/MuSiC
DENDRO (Genetic heterogeneity profiling by scRNA-seq)
https://github.com/zhouzilu/DENDRO
SAVER (Gene expression imputation and denoising for single cell RNA sequencing)
https://github.com/mohuangx/SAVER
SAVER-X (SAVER harnessing external data)
https://singlecell.wharton.upenn.edu/saver-x/
cTP-Net (Single cell Transcriptome to Protein prediction with deep neural network)
https://github.com/zhouzilu/cTPnet/
Alleloscope (Allele-specific copy number estimation for scDNA and scATAC sequencing)
https://github.com/seasoncloud/Alleloscope
Clonalscope (Subclone detection in single cell and spatial sequencing)
https://github.com/seasoncloud/Clonalscope
SCPA (Single cell pathway analysis)
https://jackbibby1.github.io/SCPA/
CellANOVA (Cell State Space Analysis of Variance, Signal recovery for single cell batch integration)
https://github.com/Janezjz/cellanova
Niche-DE (Niche-differential expression analysis in spatial transcriptomic data)
https://github.com/Kmason23/NicheDE

*For copy number profiling and tumor heterogeneity analysis using bulk sequencing:*

CANOPY (Tumor phylogeny reconstruction by spatial and temporal bulk RNA sequencing)
https://cran.r-project.org/web/packages/Canopy/
MARATHON (Comprehensive pipeline for copy number profiling in normal and tumor samples)
https://github.com/yuchaojiang/MARATHON
SWAN (Structural variant profiling using paired-end genome sequencing data)

https://bitbucket.org/charade/swan/overview

CODEX/CODEX2 (statistical framework for full-spectrum CNV profiling in whole genome, whole exome, and targeted DNA sequencing)

https://github.com/yuchaojiang/CODEX2

iCNV (Integration across array and sequencing platforms for copy number detection)

https://github.com/zhouzilu/iCNV

FALCON (Allele-specific copy number estimation using whole genome sequencing data)

https://cran.r-project.org/web/packages/falcon/index.html

FALCON-X (Allele-specific copy number estimation using whole exome sequencing data)

https://cran.r-project.org/web/packages/falconx/index.html

*General statistical tools:*

SEMBLANCE (rank-semblance kernel for data compression, niche detection, and feature extraction)

https://cran.r-project.org/web/packages/Semblance/index.html

GSEG (Change-point detection for multivariate data through a similarity graph on the observations)

https://cran.r-project.org/web/packages/gSeg/index.html

GCAT (Two-sample tests for categorical data utilizing similarity information among the categories)

https://cran.r-project.org/web/packages/gCat/index.html

SEQCBS (Segmentation and Bayesian confidence interval calculation for matched case/control point processes)

https://cran.r-project.org/web/packages/seqCBS/index.html

LEAPP (Latent factor ("batch effect") adjustment in multiple hypothesis testing)

https://cran.r-project.org/web/packages/leapp/index.html

**MEMBERSHIPS (PAST AND CURRENT)**

American Statistical Association

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ǂ *alphabetical order*
¤ *co-first authors*